Highly responsive flexible triboelectric nanogenerator using sugarcane bagasse for ultra-low pressure and vibration sensing

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https://doi.org/10.1016/j.cej.2025.167529Get rights and content

Highlights

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Highly responsive and multifunctional TENGs developed using Sugarcane bagasse (SB) biowaste and plastic waste.

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Outstanding pressure sensitivity of 188 mV/Pa in ultra-low pressure range of 1–20 Pa with detection limit of 1.32 Pa.

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A vibration sensitivity of 185 mV/g and detection of vibration frequency up to 135 Hz.

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Development of a real-time smart litter detection system using the SB-TENG.

Abstract

Vast material diversity is one of the features and possible reasons for strong research motivation in the field of triboelectric nanogenerators (TENGs). Advantages and application scope of TENGs may be widened with the incorporation of specific class of materials such as hazardous biowastes, which offer two-fold benefits in the form of energy harvesting and sustainable waste-mitigation. This study proposes the development of high-performance and multifunctional single-electrode TENGs using sugarcane bagasse (SB), a widely existing and highly polluting biowaste. Assisted by material properties of SB such as surface roughness and electro-positivity, SB-TENG produces an open-circuit voltage of ~200 V and short-circuit current of ~30 μA with a waste PET film in contact-separation operating mode. Developed from waste substances using an uncomplicated architecture and with absolutely no chemical or physical processing during the TENG preparation, SB-

TENG achieves an outstanding pressure sensitivity of 188 mV/Pa in the ultra-low-pressure regime of 1–20 Pa with 1.32 Pa as the limit of detection in comparison to its high-cost rivals developed using synthetic materials and sophisticated physico-chemical processing. As a vibration sensor, SB-TENG demonstrates a sensitivity of 195 mV/g in 0.5-10 g acceleration range and detects vibrations up to a high frequency of 135 Hz. Further, a real-time smart litter detection system developed using SB-TENG has also been presented which alerts users whenever discarded articles are thrown outside the desired trash-bin. Such studies symbolize that biowastes, which are harmful for the environment in their untreated form, can be aptly transformed not only into sustainable and environmentally-benign energy sources but also into highly-sensitive and low-cost sensors in real-time systems.

Graphical abstract

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Introduction

Research interest in the arena of triboelectric nanogenerators (TENGs) has surged rapidly. This is not only because of their merits such as simple architecture [1], outstanding electrical performance [2], and broad variety of applications [[3], [4], [5], [6], [7]], but also due to highly rich range of active materials, spanning diverse classes of synthetic [8,9] and natural materials [10,11]. Lab-made organic polymers [12,13], inorganic compounds [14], materials of animal-origin such as silk [15], animal furs [16], fish-gelatine [17], egg-shell membrane [18], human skin [19], hair [20] and plant-derived substances such as leaves [21], seeds [22], husks [23], vegetable peels [21], fruit shells [24] and coffee grounds [25] have been used for developing TENGs. This suggests their extraordinary versatility in divergent applications culminating mainly from vast series of constituent materials. Among the classes of active materials, artificial and biological waste materials obtained from different sources such as agriculture, industry and household have received significant attention, as their effective utilization in TENG development also leads to sustainable waste-mitigation besides harvesting useful electrical energy [26]. Though several categories of bio- [27] and plastic-wastes [28] have been investigated for this purpose in the earlier years, some of the common bio-wastes causing serious environmental pollution are yet to be explored for their useful potential in the energy scavenging. Further, the examination of the capabilities of bio-waste-based TENGs beyond typical energy harvesting and specifically as an integrated component of the smart sensing systems are also limited till date [28]. If meticulously undertaken, such exploration may lead to validation and promotion of energy- and environment-friendly smart IoT solutions into commercial arena.

Irrespective of the category of materials, self-powered motion sensing has been a traditional TENG application, where TENGs have been used to detect a mechanical stimulus, mainly pressure and vibration [29,30]. Plethora of studies have reported diverse materials (largely synthetic) or combination thereof, variety of physical and chemical processing methodologies and different advanced architectures of TENGs for detection of pressure and vibration [31]. The most critical challenge of such research endeavours is the optimization of structure-performance relationship, which includes optimizing the surface morphology of the active layers and overall device structure via adopting different approaches [[32], [33], [34], [35]]. These explorations have reported varied levels of performance for such sensors (pressure sensitivity reaching up to 528 mV/Pa and detection limit going low till 0.23 Pa [36]; vibration sensitivity going up to 132 V/g and maximum frequency going up till 4000 Hz [37]). However, most of these studies have explored synthetic materials and examined the sensing performance of TENGs under high magnitudes of pressure and vibration. Due to crucial challenges such as achievement of a high sensitivity, a repeatable and durable response, and involvement of simple and low-cost processing technique, studies on TENG-based sensors for ultra-low mechanical inputs are scarce till date.

Sugarcane is one of the main cash crops grown in tropical and sub-tropical regions of the world for commercial production of sugar. Remaining mass of sugarcane discarded after extracting juice is one of major agricultural and industrial biowaste, and source of detrimental air-pollution when burnt. In this article, the waste biomass obtained from sugarcane, generally known as sugarcane bagasse (SB) has been explored to develop a bio-residual-based TENG. The SB-TENG has been prepared in a single-electrode design using several other plastic waste films as tribo-negative layers in vertical contact-separation operating mode. The proposed SB-TENG generates an open-circuit voltage of ∼200 V and short-circuit current of ∼30 μA with a waste PET film. To impart flexibility, the SB-TENG has further been realized on a flexible carbon tape. The flexible architecture of SB-TENG has been demonstrated to function as high-performance pressure and vibration sensor in low-magnitude range of stimuli with remarkable performance parameters. This achievement is facilitated by utilizing SB as the active layer, which offers natural benefits such as rough and layered morphology and excellent electron-donating character of the surface functional groups, ultimately augmenting the electrical performance and sensitivity in the ultra-low-pressure regime. In addition, due to resistance to the environmental factors and high robustness, SB-TENG-based sensor demonstrates good repeatability and durability in the ultra-low-pressure regime. SB-TENG has also been incorporated as a self-powered sensor in a microcontroller-based Bluetooth-enabled real-time smart system for litter detection, which alerts the user through an android-based application on throwing trash outside the designated area. Though the utilization of SB in energy technologies signifies an alternative path to sustainable waste-mitigation like other biowastes, it explicitly encourages the innovative strategies for reducing the environmental pollution arising from specific biowastes. Further, properties such as good power density, stability and superior sensing potential of SB-TENGs have been realized with minimal usage of other resources or processing during fabrication. Therefore, the incorporation of SB using the proposed methodology simultaneously simplifies the processing and curtails the overall development cost of the device/sensor to a minimum. Such studies indicate the unexplored potential of biowastes not only as sustainable alternatives for harvesting electrical energy but also as highly-responsive low-cost sensors for detecting light-magnitude mechanical stimuli.

Materials

Sugarcane bagasse (SB) was acquired from local sugarcane processing unit while waste plastic films consisting of polyethylene terephthalate (PET), polyethylene (PE), and polypropylene (PP) were sourced from various discarded items. Carbon tape, aluminium foil and tape, polyimide (PI) tape, and double-sided adhesive tape were obtained from local vendors and chemical suppliers and used as received without any further processing.

SB-TENG assembly

SB was initially sun-dried for 4 h to eradicate any residual moisture. It was then grinded into a fine powder with a blender. The SB-TENG was assembled in a vertical contact-separation arrangement. Aluminium foil tape was attached to one side of paper-based cardboard squares (6 cm × 6 cm), with the application of carbon tape (4.8 cm × 4.8 cm) on top subsequently. SB powder was then uniformly spread over the carbon tape and hard-pressed to ensure good adhesion. Another cardboard substrate with

Results and discussion

Sugarcane (*Saccharum officinarum*) is a water-intensive crop, in the form of giant tropical grass, which is cultivated primarily for its juice. Fig. 1(a) shows the production of sugarcane bagasse as a biowaste from raw sugarcane after extracting the juice. The discarded SB has limited commercial or nutritional applications. In addition to being a major industrial biowaste, it causes severe air-pollution on burning. Therefore, sustainable alternative must be explored for its mitigation. Fig. 1(b)

Conclusion

In this article, the bio-residual sugarcane bagasse has been utilized to develop highly versatile TENGs. The TENG has been proposed in the single-electrode structure and generates a V_{OC} of ∼200 V and I_{SC} of ∼30 μA with a waste PET film in the vertical contact-separation operating mode. The flexible version of SB-TENG developed on a thin strip of carbon tape has been demonstrated to be multifunctional and highly responsive sensor to detect numerous small-magnitude mechanical stimuli. As a

CRediT authorship contribution statement

Manas Tiwari: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Deepak Bharti:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research was partially supported under the grant SRG/2020/000228 by The Science and Engineering Research Board, Department of Science and Technology, Government of India. The authors would like to acknowledge Dr. Naresh Kumar Raghuvanshi and Pradyumn Chahar for assistance in measurement of mobile vibration frequency using commercial accelerometer and Prof. S D Bharti and Dr. Jagajyoti Panda for providing assistance in vibration measurements using electrodynamic shaker system.